Why Relevance Is Not Computable

a kitchen story

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The Recognition Problem

Task: pick a cup from table, put it in the sink

But what is a "cup"?
What is a "table"?


THE RECOGNITION PROBLEM
Pattern Classification:
match against a template
category = "cup"
Serial process
Richard Duda, Pattern Classification, 2001:
Pattern recognition – the act of taking in raw data and taking an action based on the "category" of the pattern [.]…


WHAT IS A TEMPLATE?
This “template” is content:
- portable
- objective
- agent neutral
Pattern matching needs portable content
Representations!

THE FRAME PROBLEM

Frames enter the scene

Identified table, cups, sink.

frame = stereotyped situation.

1. **How to track change?** (which cup?)

2. **What is relevant?**

3. **Relevant to whom?**

The Basic Minds Problem

Cats don’t have the frame problem

They recognize the mouse is highly relevant, the moment they see one

Why?

There is no pattern classification.
Something else happens:

Pattern recognition

PRU FRAMEWORK

1. Pattern Constellation {X}

A “bundle” of patterns, **learned together**:

$$\left\{\begin{array}{l}\text{sensing patterns}\\ \text{motor patterns}\\ \text{feeling patterns}\\ \text{context patterns}\end{array}\right\}$$

PRU FRAMEWORK

2. Pattern Recognition Unity

Pattern activation and recognition are one event: experience

$$E\left(\left\{\begin{array}{l}\textbf{sensing}\ \text{patterns}\\ \textbf{motor}\ \text{patterns}\\ \textbf{feeling}\ \text{patterns}\\ \text{context patterns}\end{array}\right\}\right)$$

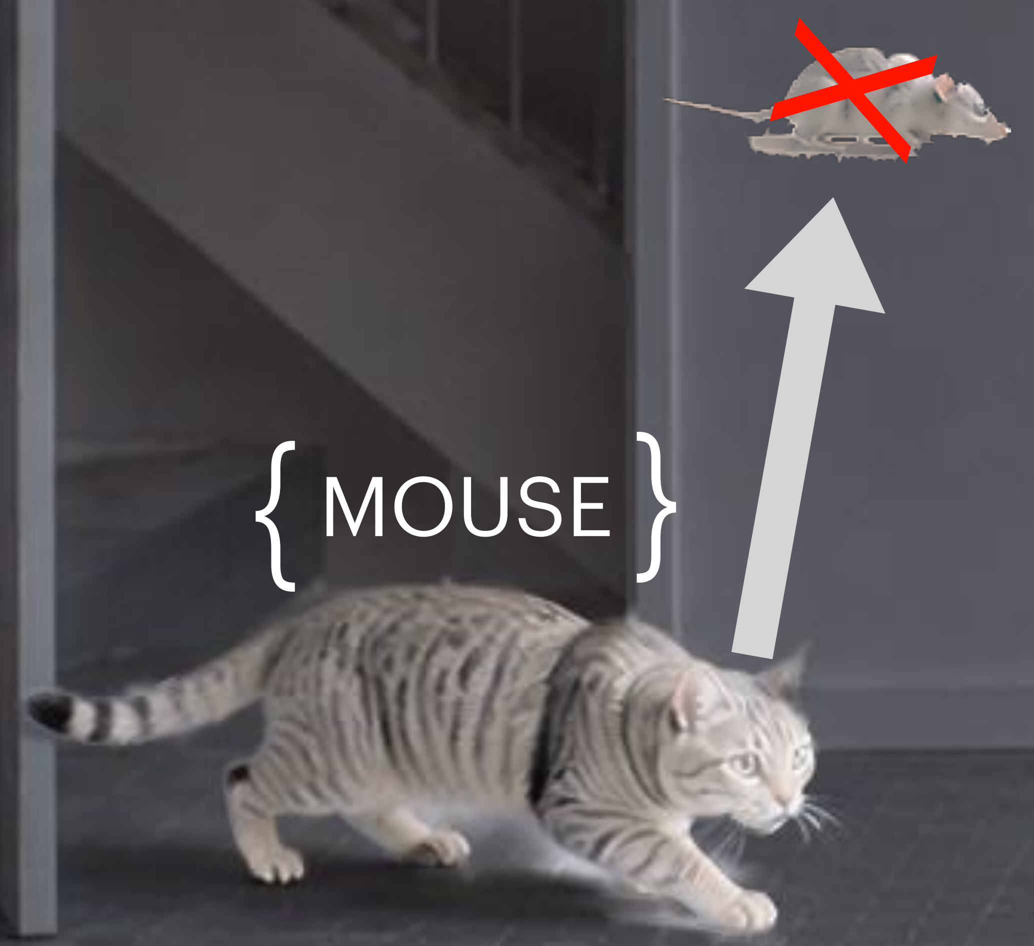
PRU: learned together, activated together

PRU FRAMEWORK

No Content, No Representation

There is no template to match,
no serial process.

{ MOUSE } is not a representation!
it is a learned NN configuration.



Dreyfus saw all of this. He argued the frame problem is what representational AI runs into, and that skillful coping sidesteps it – and by 2007 he was pointing at Freeman's attractor dynamics as the mechanism, while admitting no one had worked it out. PRU is the working-out – and it goes where Freeman didn't: connecting into language.

{X} can be modeled by a
Hopfield Network with Hebbian Learning (not chaotic!)


PATTERN CONSTELLATIONS ARE SUBJECTIVE
Descriptions (verbal) match:
- small human
- age 5
- blonde hair
- looking for a distraction
PC's (pre-verbal) don't:
cat's {MOUSE} ≠ man's {MOUSE}
{KID}
{CHILD}
{MOUSE}
{MOUSE}
{KID} ≠ {CHILD}

RELEVANCE IS LEARNED INSIDE PC's
{KID}
voice, movement
affects, read
touch, smell
quirks etc.
size,
tone of voic
generic etc.
Hubert L Dreyfus And Stuart E. Dreyfus, *How To Stop Worrying About The Frame Problem, 1987*:
For human beings, however, situations show up with relevance or salience built in, along with anticipations, based on past experience of similar meaningful situations, of how what is relevant will change as the situation develops.
Relevance is not a parameter

RELEVANCE IS SUBJECTIVE & INSTANT

WHY DO WE WANT TO COMPUTE RELEVANCE?

How does a robot takes in the world?

We supply descriptions & relevance

How Relevance Gets Stripped

Portability requires detachment from subjective value

Relevance is subjective, created in personal experience

Things turn into descriptions

Content

(representations)

Language

Symbols

How to extract relevance from language?

Frame problem

Relevance needs to be:

- pre-programmed

- extracted from language

- computed from symbols

THE LANGUAGE REALM

PRU's "word-object"

$$E\left(\left\{\begin{array}{l}\text{sensing patterns}\\ \textbf{motor}\\ \text{feeling patterns}\\ \text{context patterns}\end{array}\right\}\right) \longrightarrow cat^{o}$$

"cat!"

The Language Realm

PRU's "concept-word"

"Cats are mammals", "There was a cat here", "I want a cat", "Cat's scratch!", "The cat is out of the bag!", "She walks as a cat", etc

Language game

$\longrightarrow$ cat^c

Patterns of use

Language Pattern Constellations

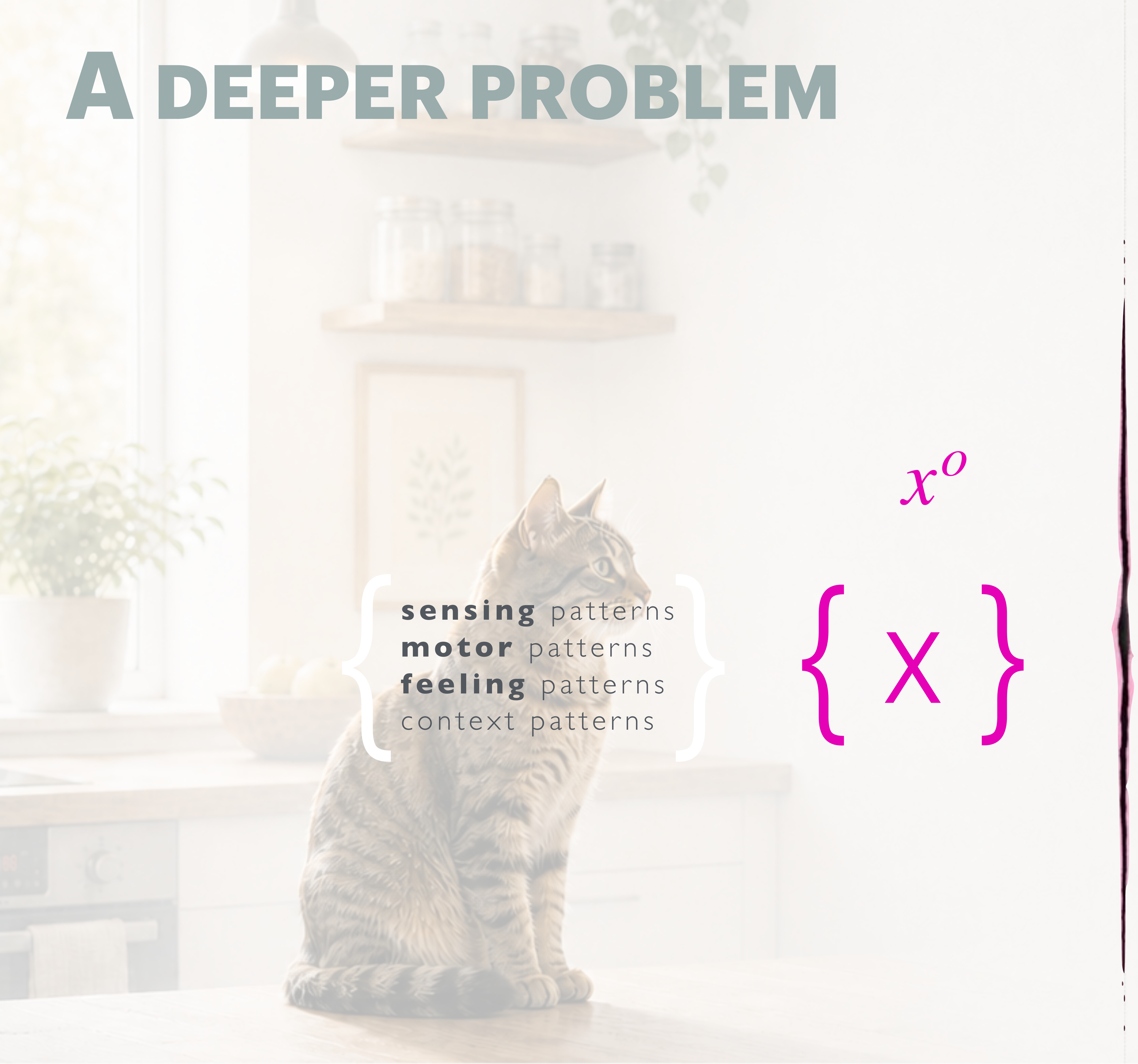
$< cat >$

Where the Frame Problem lives

Pattern Constellations

Linguistic Pattern Constellations

A DEEPER PROBLEM



sensing patterns
motor patterns
feeling patterns
context patterns

$$x^{o}$$

$$\{\, x \,\}$$

$$x^{c}$$

$$< x >$$

1. Hard Problem of Content

How does contentless basic cognition give rise to content-involving cognition?

2. Frame Problem

What is relevant so that it needs update?

3. Grounding Problem

How do a system's symbols connect to a world at all?

Basic Cognition

Language as shareable layer


Thank You!